

SHUBHAM SINGH

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SUMMARY

Forward Deployed AI Engineer who owns the end-to-end agent lifecycle — discovery, prototyping, deployment, and live monitoring and debugging — for client-facing AI/ML products. Pairs hands-on Python and React engineering with direct client collaboration to ship production-ready LLM agent systems for high-stakes support workflows.

TECHNICAL SKILLS

AI & LLMs: LLM APIs (Anthropic, OpenAI), AI Agents, Agentic Workflows, RAG Architecture, Prompt Engineering, Tool Use, LangChain, LangGraph

Languages: Python, TypeScript / JavaScript, Go, SQL

Web & APIs: React.js, FastAPI, Node.js, REST APIs, WebSockets, Async (asyncio)

Data & Retrieval: Vector Databases (Chroma, Pinecone), Embeddings, Semantic Search, PostgreSQL, Redis, MongoDB

Infra & MLOps: Docker, CI/CD (GitHub Actions), AWS, Monitoring & Observability, Model Evaluation

Ways of Working: Client Discovery, Requirement Gathering, End-to-End Ownership, Code Review, Agile / Scrum

EXPERIENCE

DataPipers — AI Engineer (Forward Deployed)

Apr 2026 – Present

Client-facing AI/ML product development

Remote, United States

- Own the end-to-end lifecycle of client-facing AI agent products — from discovery calls and requirement gathering through prototyping, deployment, and live monitoring and debugging in production.
- Lead weekly client sync meetings as the primary technical point of contact, surfacing needs, scoping solutions, and translating business requirements into shipped AI/ML features.
- Build LLM agent and Retrieval-Augmented Generation (RAG) systems in Python and FastAPI using LLM APIs (Anthropic, OpenAI) to handle client support workflows.
- Raised retrieval accuracy from 85% to 96% by introducing semantic chunking and hybrid (keyword + vector) search over client knowledge bases.
- Develop React front-end interfaces for agent interactions, delivering real-time streaming responses and a clean user experience across client deployments.
- Partner with product and core engineering teams to capture customer needs and contribute directly to the shared platform codebase.
- Monitored live production usage and cut LLM inference latency from 12s to 8s through response caching, query optimization, and migrating the serving path from Snowflake (OLAP) to an OLTP store for low-latency reads.

FreedomLedger — AI / Backend Software Developer

Jan 2025 – Jan 2026

University of Utah

Salt Lake City, UT

- Built and shipped backend services in Python (FastAPI), Go, and TypeScript / Node.js for a fintech platform, owning features from design through deployment.
- Architected a PostgreSQL database of 15+ interconnected tables with proper indexing, foreign-key relationships, encryption of sensitive data, and query optimization for sub-100ms response times.
- Integrated Plaid financial-data partner APIs with multi-layer caching to reduce redundant external calls and latency.

PROJECTS

NeuralDesk — AI Support & Knowledge Platform (RAG / LLM)

- Built an AI support agent that answers from a company knowledge base using Retrieval-Augmented Generation — Python, FastAPI, ChromaDB, Redis, and LangGraph, containerized with Docker.
- Designed multi-step agent workflows in LangGraph to route, retrieve, and resolve support queries, mirroring production high-stakes support patterns.
- Instrumented evaluation and monitoring to track answer quality and response latency across agent runs.

EDUCATION

University of Utah — B.S., Computer Science

Dec 2025

Dean's List (Spring 2021, Fall 2021)

Salt Lake City, UT

CERTIFICATIONS

Google Professional Machine Learning Engineer (PMLE) — Google Cloud, 2025.